

AI/ML Flood Risk Classification

Simplified Proof-of-Concept Technical Documentation

AI/ML Intern – Research & Product Technical Assessment

Simplified AI/ML Flood Risk PoC — Architecture & Data Flow

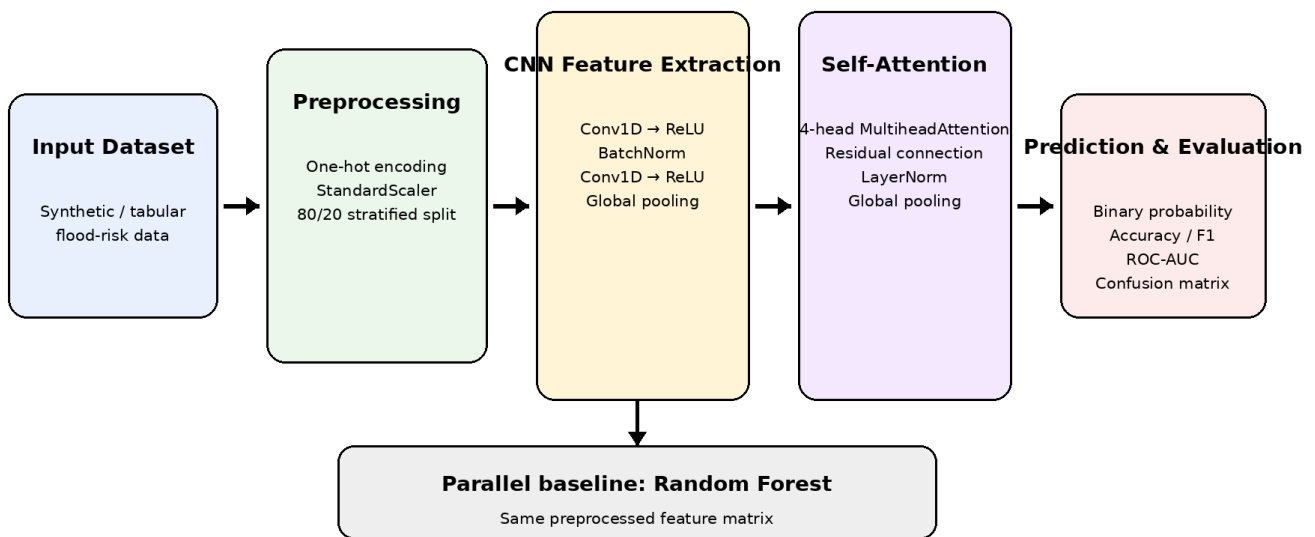


Figure 1. Implemented PoC data flow.

Executive summary. This project implements a compact flood-risk classification PoC inspired by the supplied Hydro-TransformerNet paper. The assessment permits public or synthetic data and does not require reproduction of the complete research architecture. The implementation therefore uses synthetic/tabular data rather than the original Sharjah geospatial raster pipeline.

The experiment compares a 1D CNN, a 1D CNN with multi-head self-attention, and a Random Forest baseline. The CNN + Self-Attention model demonstrates the requested CNN → Attention → Prediction concept.

1. Problem Statement and Research Context

Urban flood susceptibility depends on interacting environmental and hydrological factors. The supplied paper proposes Hydro-TransformerNet for data-scarce urban environments, combining convolutional spatial feature extraction, Transformer-based dependency modeling, and hydrologically guided attention.

The paper uses nine geospatial variables—DEM, slope, NDWI, NDVI, LULC, soil type, impervious surface, rainfall and TWI—and uses synthetic flood masks because historical flood information is unavailable. The paper reports an AUC of 0.945 in its study setup and identifies elevation, slope and rainfall as dominant factors.

PoC adaptation. This project does not claim to reproduce that architecture or result. It adapts the core learning idea to tabular data. A 1D CNN operates over the ordered feature vector and self-attention is applied to the resulting feature-position sequence.

2. Data and Synthetic Target

The dataset contains 10,000 rows and environmental/contextual variables including latitude, longitude, rainfall, temperature, humidity, river discharge, water level, elevation, land cover, soil type, population density, infrastructure and historical floods.

A new target named *Synthetic Flood Risk* was generated using a transparent hydrologically motivated score:

Factor	Weight	Effect
Rainfall	30%	Higher → higher risk
River discharge	25%	Higher → higher risk
Water level	20%	Higher → higher risk
Elevation	15%	Lower → higher risk
Historical floods	10%	Higher → higher risk

The target is synthetic and is not real historical flood occurrence.

3. Preprocessing Pipeline

Land Cover and Soil Type are one-hot encoded. The resulting experiment used 19 input features. Data was split 80/20 with stratification and a fixed seed. StandardScaler was fitted on training data and applied to the test data. The scaled matrix was converted to PyTorch tensors and reshaped to [samples, channels, features] for Conv1D.

This is intentionally simpler than the paper, which harmonizes geospatial rasters, common CRS/resolution, missing values and creates a multi-channel (C,H,W) tensor.

4. Model Architecture

CNN: two Conv1D blocks with 32 and 64 channels, ReLU, BatchNorm, adaptive global pooling and a fully connected classifier.

CNN + Self-Attention: the same CNN representation is transposed into a sequence, processed by four-head MultiheadAttention, followed by a residual connection, LayerNorm, global mean pooling and classification.

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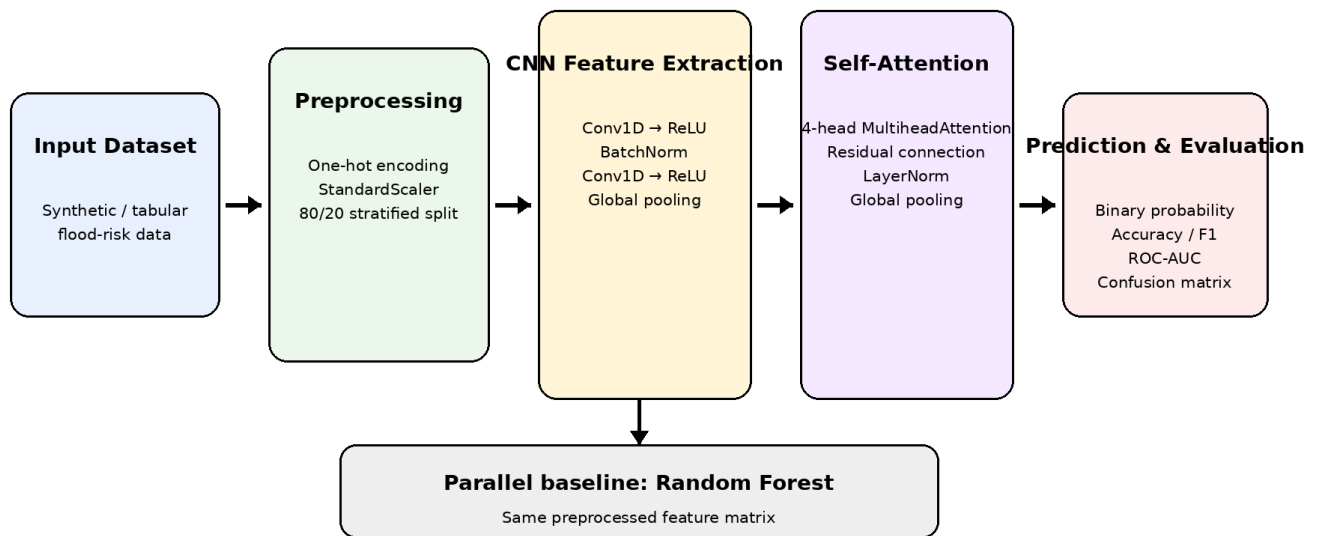


Figure 2. Architecture and parallel Random Forest baseline.

5. Training and Evaluation

Both neural models were trained for 30 epochs using BCEWithLogitsLoss and AdamW. Random Forest was trained on the same preprocessed matrix. Evaluation uses accuracy, precision, recall, F1 and ROC-AUC, with confusion matrices for error analysis.



Figure 3. Training-loss comparison from the completed experiment.

6. Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
CNN	0.7865	0.7885	0.7830	0.7858	0.8783
CNN + Self-Attention	0.7965	0.7698	0.8460	0.8061	0.8833
Random Forest	0.8285	0.8269	0.8310	0.8289	0.9153

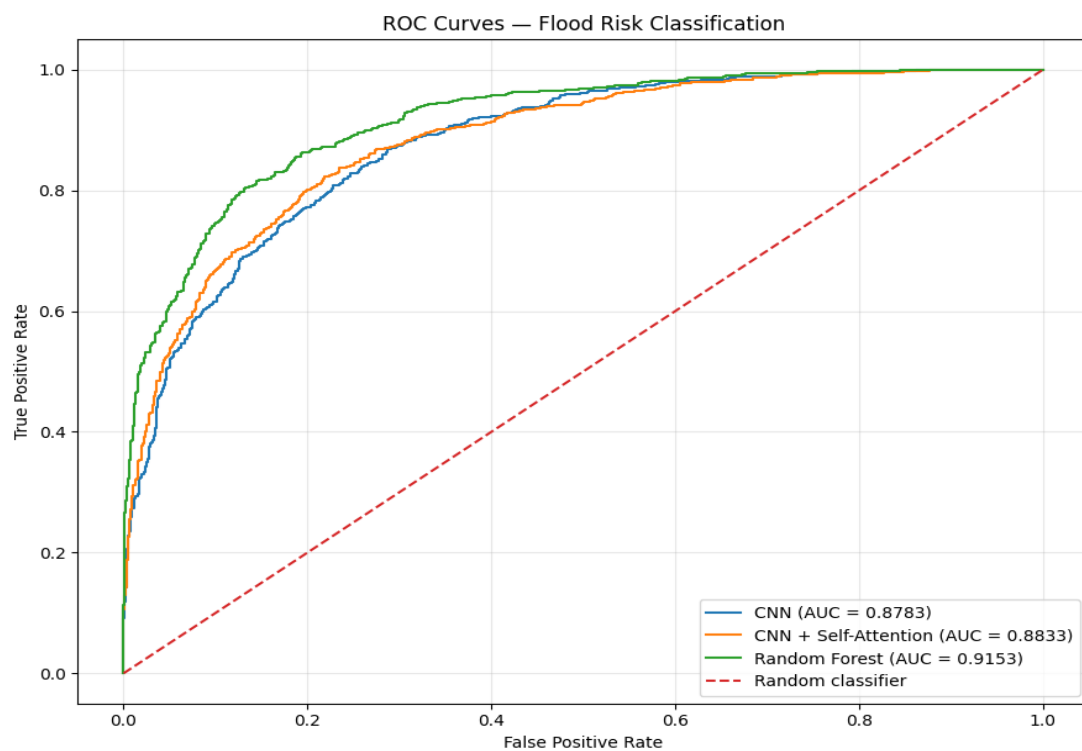


Figure 4. ROC curves from the completed experiment.

7. Error Analysis and Explainability

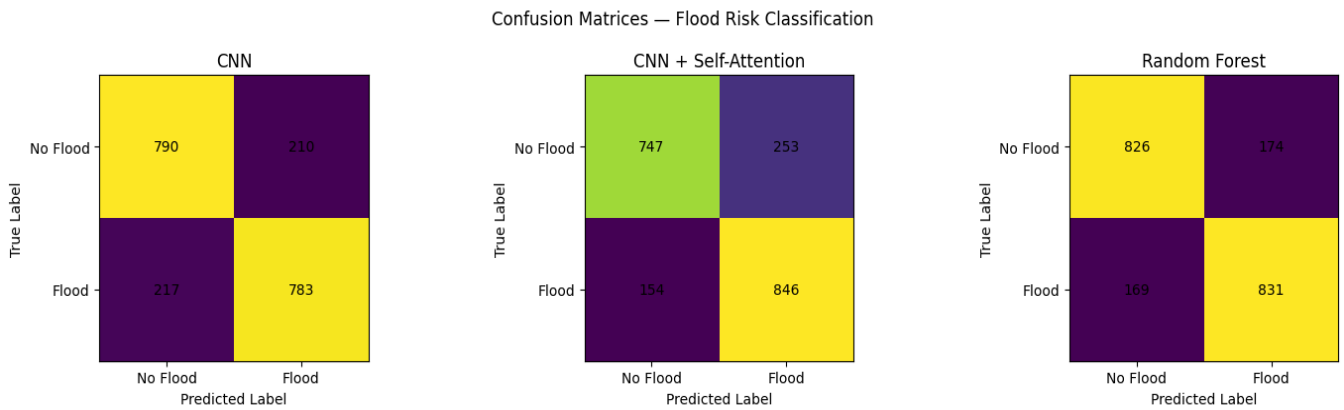


Figure 5. Confusion matrices for the three evaluated models.

CNN: TN=790, FP=210, FN=217, TP=783. CNN + Self-Attention: TN=747, FP=253, FN=154, TP=846. Random Forest: TN=826, FP=174, FN=169, TP=831.

CNN + Self-Attention increased recall relative to the plain CNN, while precision decreased. Random Forest had the highest test ROC-AUC and balanced precision/recall among the tested models.

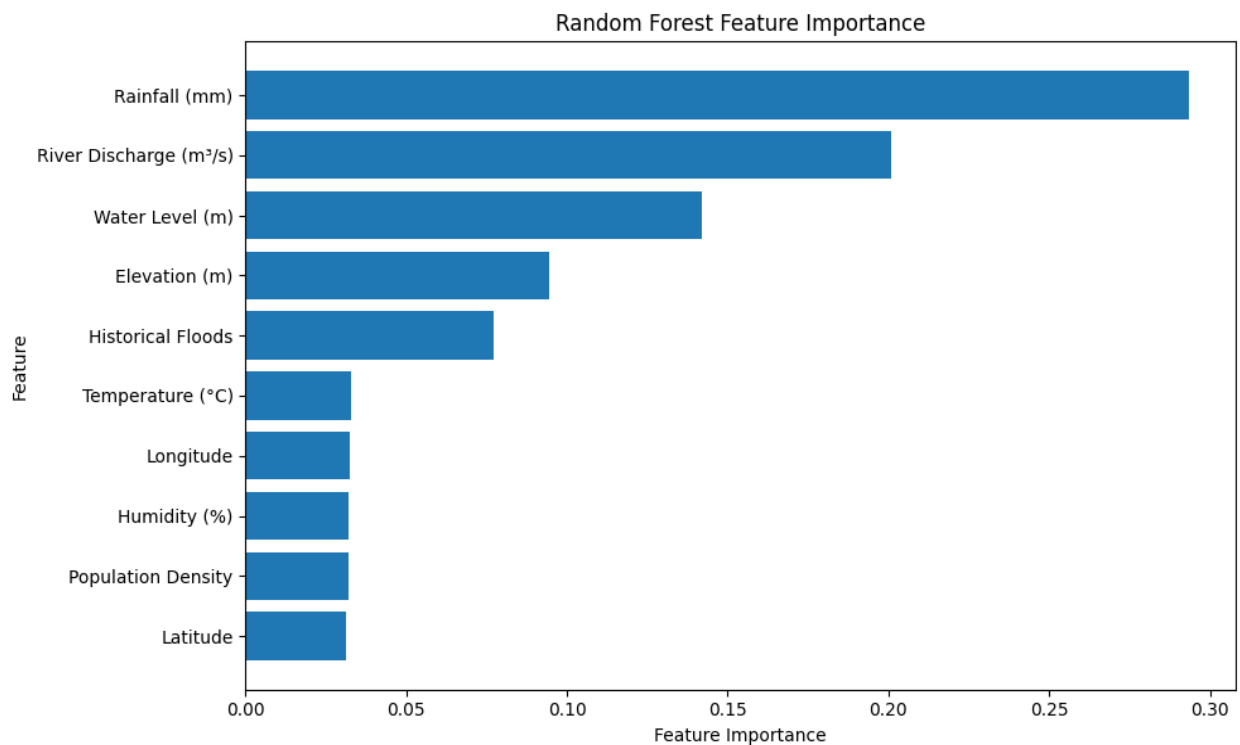


Figure 6. Random Forest feature importance. Rainfall, river discharge, water level, elevation and historical floods are the leading features.

Feature importance is model-specific and is not causal inference. In this PoC, the leading variables also participate in construction of the synthetic target.

8. Deployment, Limitations and Future Work

Deployment: a production system could expose the trained model through FastAPI. An inference endpoint would validate inputs, apply saved preprocessing, run the model and return a flood-risk probability. Production additions would include model/version tracking, logging, monitoring and a geospatial interface.

Limitations: the PoC is tabular, so Conv1D is not equivalent to a spatial 2D CNN or ResUNet. The attention block is feature-position self-attention rather than the paper's temporal Transformer. The target is synthetic, so the metrics cannot establish real-world flood-prediction performance. Random Forest feature importance is not a substitute for SHAP.

Future work: obtain genuine geospatial layers for elevation/DEM, slope, NDVI, NDWI, LULC, soil, impervious surface, rainfall and TWI; harmonize them spatially; obtain or validate flood masks; then move to a spatial encoder/ResUNet + Transformer + hydro-aware attention architecture with spatial explainability.

9. Technology Stack

Python, PyTorch, scikit-learn, pandas, NumPy, matplotlib and joblib.

10. Conclusion

The completed experiment demonstrates preprocessing, CNN feature extraction, self-attention, prediction and quantitative evaluation. CNN + Self-Attention achieved ROC-AUC 0.8833 versus 0.8783 for CNN, while Random Forest achieved 0.9153. These values apply only to this synthetic/tabular PoC and are not directly comparable with the research paper's reported AUC.

References

1. Chakraborty, R. et al. (2026). *Urban Flood Susceptibility Assessment in Arid Environment Using a Novel Hybrid Deep Learning Approach*. Earth Systems and Environment, 10, 5929–5953. DOI: 10.1007/s41748-025-00933-3.
2. Code-X-Novas AI/ML Technical Assessment, supplied assessment brief.